

Exploring **Multimodal LLMs** for Nepali ASR

A Prompting and Fine-Tuning Comparative Study

Introduction

- ▶ Automatic Speech Recognition (ASR) for Nepali remains under-served compared to high-resource languages.
- ▶ Recent multilingual multimodal LLMs (Gemini, Gemma, Qwen, Voxtral, Phi) natively accept audio input.
- ▶ They demonstrate strong multilingual speech understanding capabilities, yet lack systematic evaluation for Nepali.
- ▶ This study benchmarks 5 multimodal LLMs, 1 large pretrained ASR, and a fine-tuned Whisper baseline.
- ▶ **Emphasis:** Performance in noisy and code-switched conditions.



Motivation



General vs Task-Specific

MLLMs are used generally, but their capability on low-resource languages like Nepali is undocumented. Can they serve as practical alternatives when task-specific fine-tuning is unavailable?



Realistic Conditions

Real-world Nepali speech frequently includes noise and Nepali-English code-switching, motivating evaluation under these harder, more realistic conditions.



Intervention via Prompts

Prompt design provides a training-free intervention for studying and potentially improving low-resource ASR performance, while helping identify where fine-tuning remains essential.

Research Objectives

Primary Question: Can multilingual multimodal LLMs transcribe Nepali speech competitively—especially under noise and code-switching—and how much does fine-tuning still matter?

- ▶ **Benchmark:** 5 multimodal LLMs on Nepali ASR (zero-shot and prompted).
- ▶ **Compare:** Against Whisper Large-v3 Turbo (zero-shot) and a fine-tuned Whisper-medium at three data scales.
- ▶ **Design:** Evaluate a linguistically-aware prompting strategy based on observed Nepali ASR errors.
- ▶ **Characterize:** Identify LLM-specific failure modes (hallucination, script drift, refusal).
- ▶ **Identify:** Find the data-size crossover point where fine-tuning outperforms the best prompted LLM.

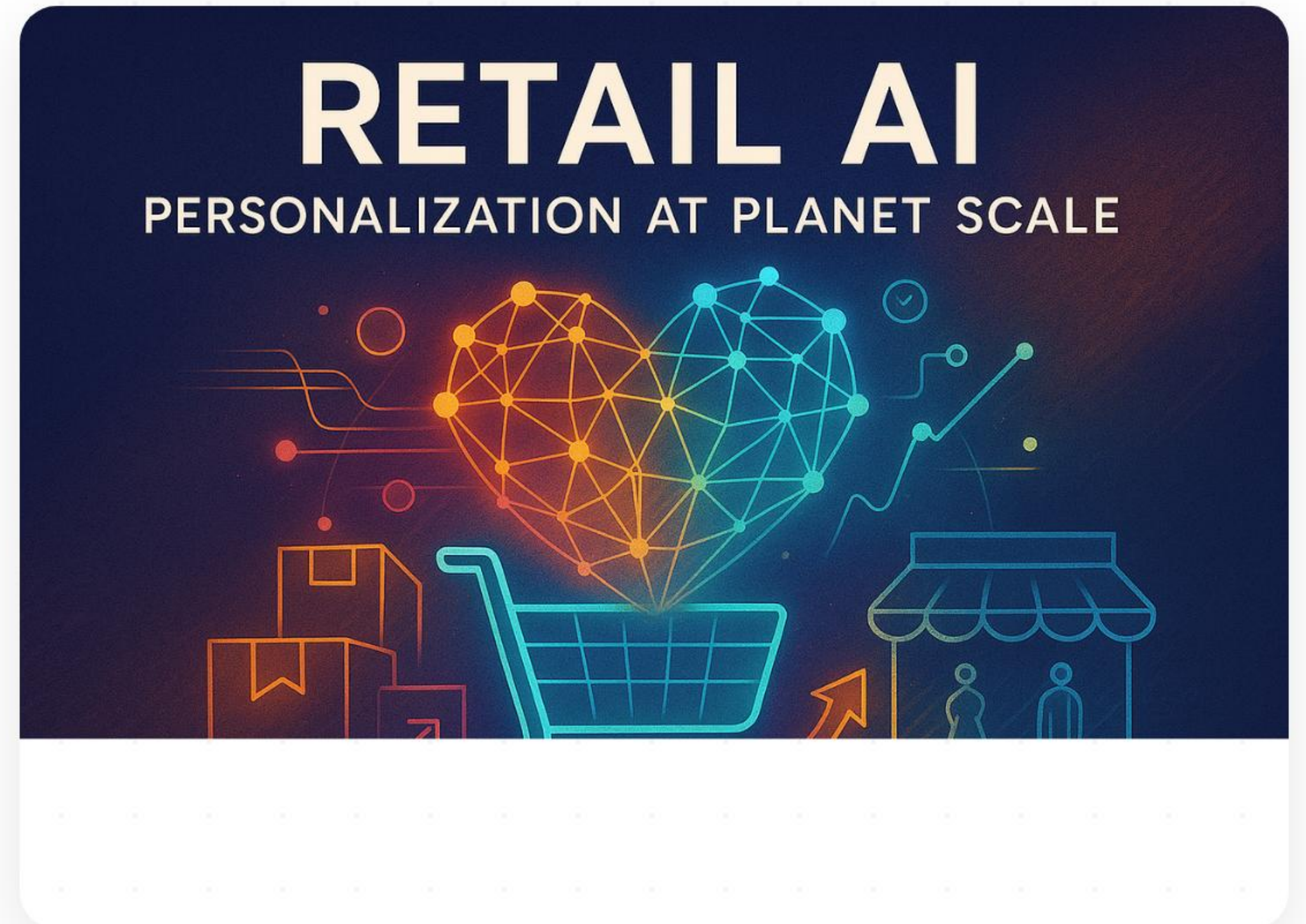
Methodology: Dataset & Models

Evaluation Set

500 utterances covering realistic conditions (Clean, Noisy, Code-switched) derived from OpenSLR54.

Candidate Models

- ▶ Gemini 3.1Pro
- ▶ Gemma 4 12B & Qwen2.5–Omni–3B
- ▶ Voxtral Mini 3B & Phi–4–multimodal
- ▶ Whisper Large–v3 Turbo (baseline)
- ▶ Fine-tuned Whisper–medium (500/2.5k/5k utterances)



Methodology: Ablation Design



Ablation Grid

- ▶ 5 Multimodal LLMs
- ▶ 3 Prompt levels: Minimal, Extensive, Linguistically-aware
- ▶ 4 Shot counts: 0, 1, 3, and 5-shot



Linguistic Prompts

The linguistically-aware prompt is decomposed into specific instructions:

- ▶ Script-related
- ▶ Morphology-related
- ▶ Code-switching
- ▶ Numeral & orthographic

Metrics & Evaluation

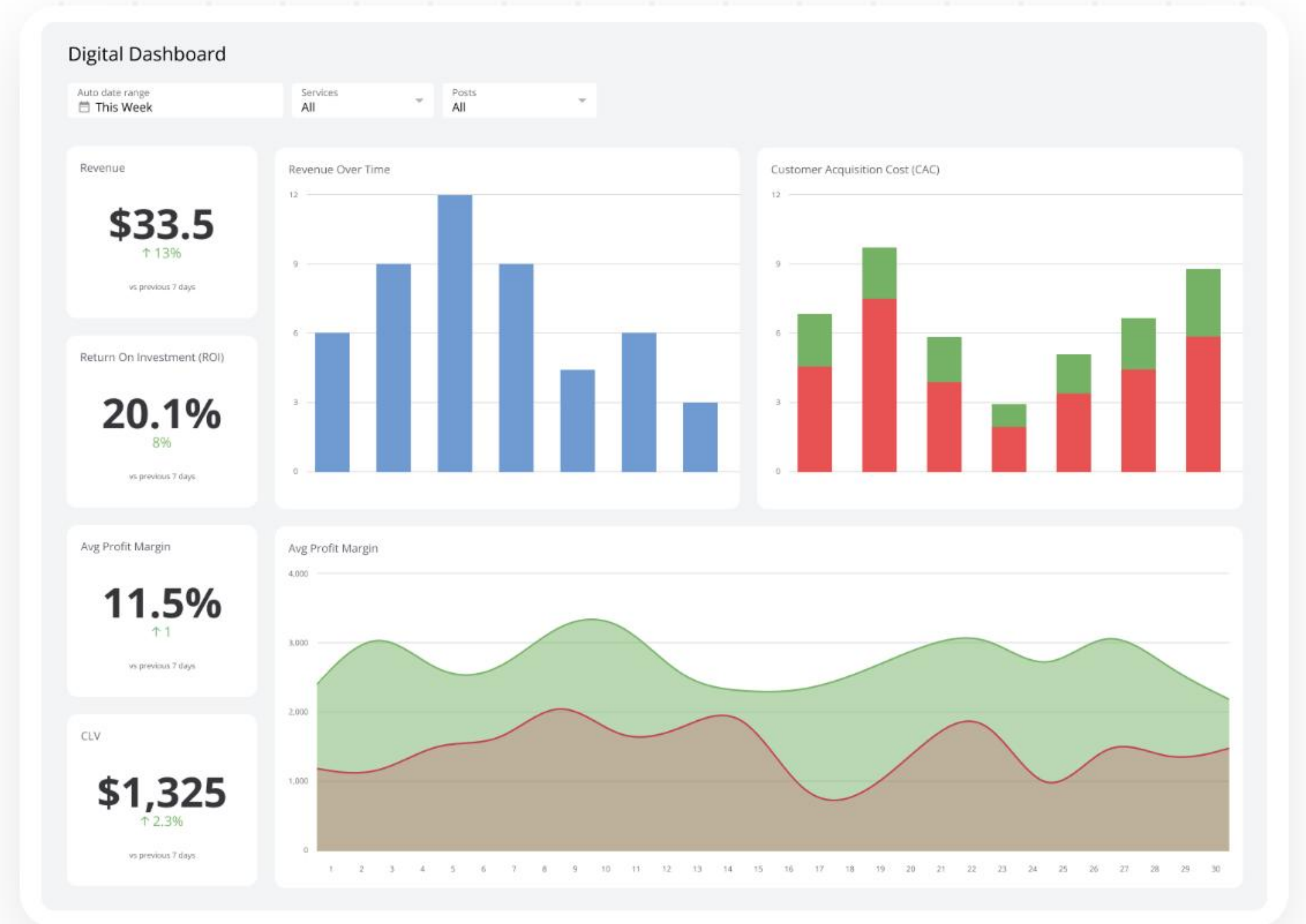
Core Metrics

Word Error Rate (WER) and Character Error Rate (CER).

LLM-Specific Failure Metrics

- ▶ Hallucination rate
- ▶ Script accuracy
- ▶ Refusal rate

Validated through manually inspected samples and tested for statistical significance. Error taxonomy will cover morphological and segmentation errors.



Automation & Feasibility



Unified Pipeline

A common interface standardizes audio input, prompting, inference, and output processing across all multimodal LLMs being evaluated.



Compute Feasibility

Experiments are carefully planned based on measured GPU runtime and memory requirements, ensuring practical execution beyond just inference counts.



Reproducibility

Open-weight models are specifically selected to run on freely available Kaggle T4/P100 GPUs, enabling replication without needing paid compute resources.

Project Timeline (7 Weeks)

Week 1

Dataset preparation and pipeline setup

Week 2-3

Benchmark experiments and model fine-tuning

Week 4

Evaluation and statistical analysis

Week 5

Error analysis and prompt ablation

Week 6-7

Paper writing and revision

Expected Outcomes

Category	Expected Delivery / Insight
Systematic Evaluation	First systematic evaluation of MLLMs on Nepali ASR, focusing on realistic noisy and code-switched conditions.
Prompting Framework	A linguistically-aware framework addressing script, morphology, and code-switching challenges.
Empirical Crossover	Defining the exact data-size needed where fine-tuning outperforms the best zero-shot LLM.
New Metrics	Validated failure-mode metrics (hallucination, script fidelity) for future low-resource evaluations.
Practical Advice	Deployment recommendations for teams operating without resources for fine-tuning.

Thank you

Questions?

Image Sources



<https://www.bgr.com/img/gallery/9-incredible-high-tech-hearing-aids-features-that-help-people-hear-again/directional-microphones-1780593070.jpg>

Source: www.bgr.com



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Source: medium.com



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